

Complete Architecture, Working Process, Business Rules & Development Governance Discovery

Complete 40-Section Authoritative Systems Specification & Operating Manual

Next.js 16.3.0 App Router

React 19.2.8

Supabase PostgreSQL 15+

TypeScript 5

Tailwind CSS v4

Row Level Security (RLS)

DOCUMENT TYPE Architecture Master Document (Full 40 Sections)	VERIFICATION STATUS 100% Codebase Verified (Read-Only Discovery)
SECURITY & TENANCY Next.js 16 Proxy + Server Action Gateways + PostgreSQL RLS	AUDIENCE Senior Developers, Software Architects & Engineering Teams

1. Executive Summary

The **Roshani Public School ERP** is an enterprise school management system designed for robust academic administration, multi-tenant security, and double-entry financial governance. Built on Next.js 16 and Supabase PostgreSQL, it enforces two separate hierarchies: **System Authority** and **School Authority**.

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Browser → Next.js 16 Proxy (updateSession) → Server Layout (resolveUser) → RBAC (authorizeAction) → Supabase RLS (sch
```

2. Technology Stack

Layer	Technology	Version	Purpose
Frontend / Server	Next.js (App Router)	16.3.0	SSR, Server Actions, Next.js 16 Proxy
UI Runtime	React / React-DOM	19.2.8	Server components, transitions, React.cache()
Language	TypeScript	^5.x	End-to-end type safety
Database	PostgreSQL (Supabase)	15+	Relational DB, stored procedures, RLS
Auth & Session	@supabase/ssr / GoTrue	^0.12.4	Cookie-based SSR session management
Styling	Tailwind CSS	v4	HSL design tokens & responsive components
Validation	Zod	^4.4.3	Authoritative server & client validation
Form Handling	React Hook Form	^7.85.0	Controlled & uncontrolled form state
Testing	Vitest / Playwright	^4.1.10 / ^1.62.1	Unit, integration, and E2E testing
Payments	Razorpay Webhooks	HMAC SHA-256	Idempotent webhook reconciliation

3. Repository Architecture

The repository follows a clean domain-oriented modular layout:

- src/proxy.ts : Next.js 16 proxy layer replacing legacy middleware.
- src/app/erp/ : 6 role portals (admin, principal, teacher, accountant, student, parent).
- src/app/verify/ : Public token-indexed verifiers for admit cards, certificates, and report cards.
- src/lib/ : Domain business engines (auth, admissions, attendance, examinations, fees, leave, notifications, students).
- supabase/migrations/ : 39 sequential database migration SQL scripts.

4. Application Routing Architecture

Portal	Route Pattern	Roles	Scope & Data Access
Root Dispatcher	/erp	Authenticated	Redirects to primary role portal or /erp/select-role
Admin	/erp/admin/*	Super Admin, Admin	School-wide operations (admissions, attendance, exams, fees)
Principal	/erp/principal/*	Principal	Academic governance, marks approval, results publish, certificates
Teacher	/erp/teacher/*	Teacher	Scoped strictly to assigned classes, sections, and subjects
Accountant	/erp/accountant/*	Accountant	Fee collection cash register, receipts, daily cash reconciliation
Student	/erp/student/*	Student	Read-only self access to personal attendance, results, fees
Parent	/erp/parent/*	Parent	Multi-child dashboard scoped to verified linked children
Public Verifier	/verify*/[token]	Public (No Auth)	Token-indexed snapshot verification (admit card, cert, result)

5. User Roles & RBAC Architecture

The system separates **System Authority** (Super Admin, Admin) from **School Authority** (Principal, Vice Principal, Coordinator, Teacher, Student, Parent).

Feature	Super Admin	Principal	Admin	Teacher	Accountant	Student	Parent
Platform Config	Full	View	View	No	No	No	No
Admissions Convert	Yes	Yes	Yes	No	No	No	No
Attendance Mark	Yes	Yes	Yes	Assigned	No	No	No
Fee Structure Approve	Yes	Exclusive	No	No	No	No	No
Fee Collections	Yes	View	Yes	No	Full	No	No
Admit Card Override	Yes	Exclusive	No	No	No	No	No
Marks Approval	Yes	Exclusive	No	No	No	No	No
Results Publish	Yes	Exclusive	No	No	No	No	No
Certificate Issue	Yes	Yes	Yes	No	No	No	No
Staff Leave Approve	Yes	Exclusive	No	No	No	No	No

6. Authentication & Session Architecture

Operates through a 5-step pipeline: (1) Supabase GoTrue credentials validation → (2) In-memory rate limiting with progressive delay & temporary lockout → (3) Next.js 16 Proxy session cookie refresh via `updateSession()` → (4) Server identity resolution (`resolveUser()`) cached per request → (5) Multi-tab logout broadcast via `BroadcastChannel('rps_erp_auth_sync')`.

7. Security Architecture

Enforces multi-tier defense: (1) Server-side identity validation, (2) Centralized RBAC evaluator (`rbac.ts`), (3) Row Level Security (RLS) in PostgreSQL, (4) Strict open-redirect defense (`redirect.ts`), (5) Brute-force throttling (`rate-limiter.ts`), (6) HTTP Security Headers (CSP, HSTS, X-Frame-Options), (7) Razorpay HMAC SHA-256 webhook verification.

8. Multi-Tenant Architecture

Tenant isolation is anchored at the database level by the `school_id` foreign key across 40+ tables. Helper functions `public.get_current_school_id()` and `public.has_role()` filter all queries. Server actions derive `schoolId` exclusively from the authenticated user profile, never trusting client parameters.

9. Database Architecture

Core entity model contains 40+ relational tables: `schools`, `school_settings`, `academic_sessions`, `classes`, `sections`, `subjects`, `class_subjects`, `profiles`, `roles`, `user_roles`, `guardians`, `students`, `student_guardians`, `student_academic_history`, `student_documents`, `admission_applications`, `audit_logs`, `teacher_assignments`, `attendance_sessions`, `attendance_records`, `leave_types`, `leave_applications`, `leave_approvals`, `leave_entitlements`, `notifications`, `fee_heads`, `fee_structures`, `fee_structure_items`, `student_fee_assignments`, `student_concessions`, `invoices`, `invoice_items`, `payments`, `payment_allocations`, `financial_ledger`, `refunds`, `adjustments`, `receipts`, `daily_cash_reconciliations`, `cash_movements`, `exam_types`, `examinations`, `examination_schedules`, `examination_subject_configs`, `admit_cards`, `grading_scales`, `student_marks`, `student_results`, `promotion_policies`, `class_progressions`, `promotion_records`, `report_cards`, `certificate_types`, `certificate_sequences`, `certificates`.

10. Database Migration Governance

All database changes are managed via 39 sequential migration files in `supabase/migrations/` named with `YYYYMMDDHHMMSS_<description>.sql`. Migrations are strictly additive and backward-compatible. Manual production schema

modifications are forbidden.

11. Server / Client Architecture

Client components (`'use client'`) handle user interaction, form inputs, and optimistic UI. Server components and Server Actions (`'use server'`) represent the authoritative security boundary performing user resolution, RBAC validation, audit logging, and database execution.

12. Server Action Rules

Every sensitive Server Action follows an immutable 9-step execution sequence: Session Authenticate → RBAC Verify → Zod Schema Validate → Tenancy Derive → State Machine Check → Database Mutation → Audit Log Write → Cache Revalidate → Safe Structured Response.

13. Validation Architecture

Validation occurs at three tiers: Client-side (React Hook Form for instant feedback), Server-side (Zod schemas as authoritative gatekeeper), and Database-level (foreign keys, NOT NULL, and CHECK constraints for absolute integrity).

14. Admissions Workflow

Workflow: `draft` → `submitted` → `under_review` → `approved` → `converted` . Atomic conversion executes stored procedure `convert_admission_application` to create student records, link guardians, and register initial academic history.

15. Student Workflow

Covers enrollment, class assignment, section changes, guardian linking, and academic history tracking. Students have self-access; Parents access verified linked children via `student_guardians` .

16. Attendance Workflow

Teachers submit daily attendance for assigned sections. Modifying a submitted session strictly requires a `correctionReason` of at least 3 characters. Administrators and Principals can lock sessions permanently.

17. Examination Architecture

Enforces canonical exam terminology: `UT` (Unit Test), `PT` (Periodic Test), `HY` (Half-Yearly Examination), `PA` (Pre-Annual), `ANNUAL` (Annual Examination), `PRAC` (Practical), `IA` (Internal Assessment).

18. Admit Card Workflow

Generates hall tickets after verifying financial clearance (or Principal fee hold override). Creates a 256-bit cryptographically secure token, a document fingerprint (`RPS-AC-{YEAR}-{HASH}`), and an immutable data snapshot. Reissuance sets previous versions to `superseded` .

19. Results Workflow

Marks progression: `draft` → `submitted` → `pending_approval` → `approved` → `published` → `locked` . The deterministic calculation engine evaluates totals, percentages, and resolves letter grades from `grading_scales` without altering enrollment.

20. Fees Workflow

Fee structures require Principal approval before activation. Double-entry financial ledger postings accompany all invoice settlements. Refund transactions are protected by row-level locking (`FOR UPDATE`) to prevent concurrent double-refunding.

21. Leave Workflow

Student/Parent leave is reviewed and approved by Admin or Principal. Staff/Teacher leave is exclusively approved or rejected by the Principal. Leave usage is tracked against annual allowances in `leave_entitlements`.

22. Promotion Workflow

Evaluates year-end results against `promotion_policies` and `class_progressions`. Decisions: `PROMOTED`, `PASSED_OUT`, `SUPPLEMENTARY`, `CONDITIONAL_PROMOTION`, `REPEAT`. Promoted students receive new history records in target sessions.

23. Documents Workflow

Supports Report Cards and official Academic Certificates (`TC`, `BONAFIDE`, `CHARACTER`, `COMPLETION`, `STUDY`). Certificate numbers are atomically generated via `generate_certificate_number` RPC in the format `RPS/PREFIX/YEAR/000001`.

24. Notification Architecture

Database-backed in-app notification engine (`notifications` table) with real-time unread counter badge in header and deep links to relevant administrative modules.

25. Audit Logging

All sensitive operations (login, logout, admissions, attendance, fees, marks, results, certificates) write immutable records to `audit_logs` capturing actor ID, school ID, entity, and old/new JSON payloads without storing sensitive secrets.

26. UI/UX Governance

Clean enterprise slate palette (`bg-slate-50`, `text-slate-900`, `border-slate-200`) with deep blue brand accents (`bg-blue-600`). Tabular monospaced figures (`font-mono`) for IDs, admission numbers, fingerprints, and currency.

27. Selection Control Rules

In cascading dropdowns (Session → Class → Section → Student), changing a parent selection immediately resets dependent child selections to empty string `""` to prevent invalid cross-entity form submissions.

28. Error Handling Conventions

Sanitizes all errors into human-readable user messages using the standard `ActionResult<T>` envelope. Never exposes PostgreSQL codes, connection strings, table names, or server stack traces.

29. Performance Architecture

Server-side data fetching in Server Components, `React.cache()` request deduplication for auth resolution, compound database indexes on (`school_id`, `class_id`, `section_id`), and optimized package imports for icons.

30. Testing Architecture

Unit & integration testing via Vitest covering RBAC, permissions, dropdowns, and calculation engines. End-to-end browser testing via Playwright covering multi-role login, admissions, fees, and marks pipelines.

31. Non-Negotiable Development Rules

RULE-001	ZERO CLIENT TRUST
Never trust client-supplied <code>school_id</code> , <code>profile_id</code> , or <code>roles</code> . All identity must be derived server-side via <code>resolveUser()</code> .	
RULE-002	SERVER-SIDE AUTHORIZATION
Every Server Action must invoke <code>resolveUser()</code> and <code>authorizeAction()</code> before performing any database query or mutation.	
RULE-003	ROW LEVEL SECURITY MANDATE
Every table in PostgreSQL must have Row Level Security enabled with <code>school_id = public.get_current_school_id()</code> .	
RULE-004	IMMUTABLE FINALIZED RECORDS
Approved fee structures, issued certificates, and locked results cannot be overwritten in place; changes require formal versioning or controlled audit unlock.	
RULE-005	SELECTION CONTROL CASCADING
In multi-tier select controls (e.g. Session → Class → Section → Student), changing a parent selection must immediately reset all dependent child values.	
RULE-006	ADDITIVE DATABASE MIGRATIONS
Schema changes must be added as sequential migration scripts in <code>supabase/migrations/</code> . Direct manual production edits are strictly prohibited.	

32. Change Management Process

Follows standard lifecycle: Understand Requirement → Inspect Architecture → Verify Authorization → Implement Minimal Changes → Run Unit Tests (`npm test`) → Type Check (`tsc`) → Build Validation (`npm run build`) → Security Review.

33. Protected Architectural Components

Foundational modules that must not be altered without senior review: `resolve-user.ts`, `rbac.ts`, `permissions.ts`, `workflow.ts`, `src/proxy.ts`, `rate-limiter.ts`, RLS helper functions, and calculation engines.

34. Known Limitations

In-memory rate limiter is instance-local (requires Redis/KV for multi-node serverless clusters); Razorpay webhook requires live environment secrets in production; background scheduled cron tasks require external runner (e.g. `pg_cron` or Vercel cron).

35. Current Maturity Assessment

Architecture: **Excellent** | Security: **Excellent** | Database: **Excellent** | Business Workflows: **Excellent** | Testing: **Good** | Performance: **Good** | UI/UX: **Good** | Documentation: **Excellent**.

36. How This ERP Works

A user logs in → GoTrue validates JWT → `resolveUser()` derives profile, school, and roles → Next.js 16 Proxy refreshes session → Server layout mounts `ERPAppShell` → Server Actions enforce RBAC, mutate PostgreSQL, and write audit logs.

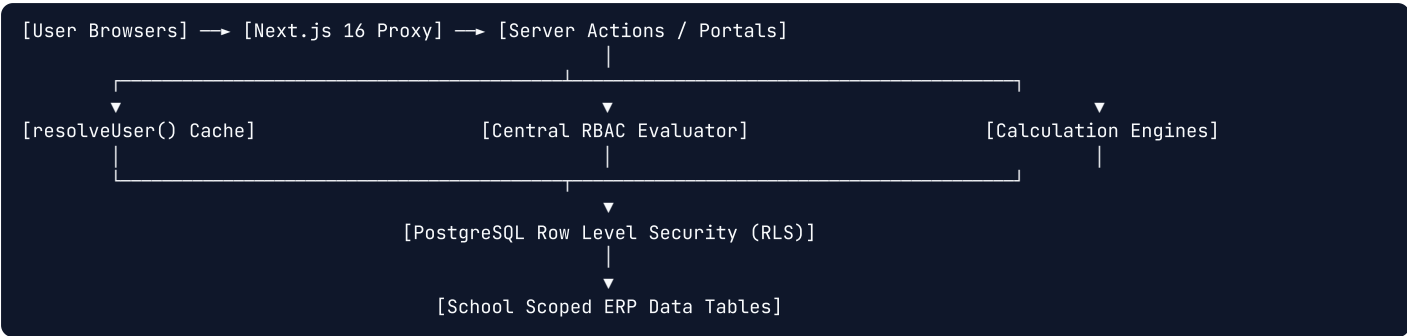
37. Permanent Rulebook

Summarizes governance invariants: `RULE-001 (Tenant Isolation)`, `RULE-002 (Server Authorization)`, `RULE-003 (State Machine Integrity)`, `RULE-004 (Canonical Terminology)`, `RULE-005 (Audit Logging)`, `RULE-006 (Selection Cascading)`.

38. Developer Onboarding Guide

5-Day Plan: Day 1 (Architecture & Portals) → Day 2 (Auth, RBAC & RLS) → Day 3 (Database & Core Workflows) → Day 4 (UI Components & Testing) → Day 5 (Deployment & Security Review).

39. Architecture Diagram



40. Final Recommendations

- 1. Strictly uphold server-side authorization boundaries;
- 2. Preserve sequential additive migrations in `supabase/migrations/`;
- 3. Maintain full Vitest and Playwright test coverage for all new workflows;
- 4. Adhere to canonical examination terminology.